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Review

Complications of vertebral augmentation in metastatic spinal fractures: a systematic review of multiple study designs

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HIGHLIGHTS

- Largest review of vertebral augmentation complications in metastatic disease.
- Symptomatic complications were uncommon (3.4%) despite frequent cement leakage.
- CT detected more cement leakages with no change in outcome (most are asymptomatic).
- Rare but serious complications were identified through case reports.
- The apparent safety profile depends on how complications are defined and detected.

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ABSTRACT

Background: Vertebral augmentation techniques, including vertebroplasty and kyphoplasty, are widely used to relieve pain and restore spinal stability in patients with metastatic vertebral lesions. Despite their widespread use, the full spectrum and clinical significance of procedure-related complications remain incompletely characterised. This systematic review aims to summarise and characterise all reported complications associated with vertebral augmentation for metastatic thoracolumbar disease.

Methods: A comprehensive search of PubMed/MEDLINE, Scopus, and Web of Science databases was performed. Studies reporting complications associated with vertebral augmentation procedures for metastatic thoracolumbar spinal fractures were included. Randomised controlled trials, observational studies, case series, and case reports were eligible. Due to substantial heterogeneity across studies, a descriptive synthesis was performed.

Results: A total of 76 studies were included, comprising 3234 patients and at least 5737 vertebral levels treated. The overall weighted complication rate was 25.5%, while the weighted rate of symptomatic complications was 3.41%. Cement leakage was the most frequently reported complication, with a weighted mean incidence of 20.7%, although the majority were asymptomatic. Other complications included postprocedural pain (1.77%), pulmonary cement embolism (1.08%), and adjacent level fractures (0.79%). Studies using CT imaging reported significantly higher cement leakage rates (28.6% vs 11.9%, $p = 0.004$). Case reports described rare but potentially severe complications, including needle-tract tumour seeding, tumour extrusion, and cement migration to cardiac structures.

Conclusions: This review presents a comprehensive and detailed description of all complications of vertebral augmentation for treatment of metastatic spine disease and reinforce its safety. Differences in imaging protocols and reporting standards contribute to variability in reported complication rates. Further prospective studies with standardised complication reporting are needed.

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1. Introduction

Advances in oncological therapies have improved survival rates and consequently increased the prevalence of metastatic disease and its complications, with 70% of patients developing spinal metastases [1,2]. Metastatic involvement of the spine disrupts normal bone architecture, increasing its susceptibility to vertebral compression fractures (VCFs), associated with axial pain, mechanical instability, spinal deformity, and even neurological compromise. These manifestations may impair mobility, quality of life and increase analgesic use and hospitalisation [1,2]. Treatment strategies for management of metastatic spinal disease are mainly palliative and aim to control pain, maintain or restore spinal stability, preserve neurological function, and optimise quality of life [1,3].

Minimally invasive vertebral augmentation techniques, as percutaneous vertebroplasty and balloon kyphoplasty, have emerged as important treatment options for painful cancer-related VCFs [1,4]. These procedures involve the percutaneous injection of polymethylmethacrylate (PMMA), into the fractured vertebral body, improving mechanical stability and load-bearing capacity [5]. Multiple studies have shown that vertebral augmentation provides effective pain control, reduces opioid consumption, and improves functional outcomes. [1,4,6,7] . These benefits are particularly relevant in palliative care, where rapid symptom control and preservation of autonomy are essential. [1,3].

The increasing use of vertebral augmentation procedures has drawn attention to a range of perioperative and postprocedural complications. These complications vary from minor and self-limiting events to rare but potentially severe and life-threatening outcomes. Cement leakage is the most frequently reported complication, with incidence rates varying widely across studies and often exceeding 20% [5,8]. Although most cases of cement leakage are asymptomatic, it may result in clinically significant consequences, such as nerve compression, pulmonary and intracardiac embolism, adjacent vertebral fractures, and local pain, highlighting the need for careful procedural planning [8]. Recent systematic reviews have attempted to identify risk factors for cement-related complications, but findings remain inconsistent. [8] Other reported complications include infection, paravertebral haemorrhage or haematoma, pedicle fractures, allergic reactions to bone cement, postoperative pain, and deep vein thrombosis. [5,9]. These events are uncommon and often described only in case reports or small case series.

This systematic review aims to comprehensively evaluate both common and rare complications associated with vertebral augmentation for metastatic spinal disease, and to clarify their incidence, risk factors, and clinical relevance.

2. Methods

2.1. Study design

A systematic review including multiple study designs (SR-MSD) was conducted to provide a comprehensive overview of complications associated with vertebral augmentation for metastatic spinal disease. Randomised controlled trials (RCTs), observational studies, case series, and case reports were included to capture both common and rare complications. Despite their lower level of evidence, case reports were included as they may provide valuable insights in oncologic spine care, where high-quality trials remain limited.

2.2. Search strategy and research questions

This systematic review was conducted in accordance with the PRISMA guidelines [10]. A systematic literature search was performed in PubMed/MEDLINE, Scopus, and Web of Science from database inception to July 2025. The search strategy combined terms related to metastatic spinal disease, vertebral augmentation procedures (including vertebroplasty and kyphoplasty), and postoperative complications. The complete search strategies for each database are provided in Appendix A.

2.3. Study selection and eligibility criteria

All records identified through the database search were imported into a reference management software, and duplicates were removed. Study selection was conducted independently by two reviewers (MCC and AMV), through title/ abstract screening followed by full-text review. Disagreements between reviewers were resolved by a third reviewer (MJL). No restrictions were imposed regarding year of publication or language.

The inclusion criteria were patients with thoracolumbar spinal metastatic disease submitted to vertebral percutaneous augmentation procedures (vertebroplasty, kyphoplasty); studies reporting data on complications; RCTs, observational clinical studies (cohort studies, case series, case–control studies) and case reports; and investigations conducted in a clinical setting.

The exclusion criteria were: primary bone tumours or osteoporotic fractures; vertebral augmentation of cervical (different surgical approach) or sacral spine (irregular shape); neurological compromise before surgery; vertebral augmentation techniques performed simultaneously with other spinal interventions (surgical fixation, bone tumour ablation procedures, conjoined use of bone filling mesh, or others); and systematic reviews, meta-analyses, scoping reviews, abstracts, editorials, book chapters, conference papers, commentaries, opinion articles, and narrative reviews without original data. If complications attributable exclusively to vertebroplasty or kyphoplasty were clearly documented, the articles were included, and only those patients were considered in data extraction. Case reports describing complications related to percutaneous vertebroplasty or kyphoplasty were included, whereas reports focusing exclusively on procedural benefits or efficacy without mention of complications were excluded.

In comparative studies (RCTs, case-control studies, and two-arm cohort studies), data extraction was tailored to the relevance of the comparison groups to the outcome of this review. When appropriate, only the vertebroplasty or kyphoplasty treatment arms were included, whereas in studies where the comparison was not relevant to the evaluation of complications, the overall study population was considered. As the primary objective was to characterise complications rather than compare interventions, all studies were analysed descriptively. In one study, both arms were analysed separately and treated as independent series, once the study specifically compared vertebroplasty and kyphoplasty [11].

The absence of reported complications in some studies was interpreted as not reported rather than the absence of events, and those studies were therefore excluded.

2.4. Data extraction and synthesis

Data extraction was independently performed by two reviewers (MCC, AMV) using a standardised data extraction form. Any discrepancies in data extraction were resolved through discussion and consensus between the reviewers.

The following variables were collected: study characteristics (author, year, country, study period, design); patient characteristics (number of patients, treated levels, age, tumour sites, distribution and characteristics of involved levels); treatment characteristics (type of vertebral augmentation procedure, anaesthesia, fluoroscopy/CT guided, volume of bone cement, number of treated vertebras per session); reported postoperative complications (frequency, severity, risk factors; leakage type and symptomatic events).

Due to significant heterogeneity of the studies included information a quantitative meta-analysis was not feasible, and a descriptive synthesis of complications was performed.

2.5. Risk of bias assessment

Methodological quality was evaluated using the Joanna Briggs Institute (JBI) [12] critical appraisal tools according to study design (JBI checklist for randomised controlled trials, cohort studies, and case-control studies). Case series and case reports were not formally assessed for publication bias, as their study designs are considered inherently susceptible to selection and reporting bias.

3. Results

3.1. Study selection

After a full electronic search, a total of 1224 references were identified. After duplicate removal, 682 records were excluded based on title and abstract screening. 134 full text articles were accessed for eligibility, of which 70 full text records were excluded according to the exclusion criteria, leaving 59 articles to be included finally. 17 additional studies were identified following backward citation. In total, 76 articles were included in this review. The study selection process is illustrated in the flow diagram in Fig. 1.

3.2. Characteristics of included studies

Most of the included studies consisted of descriptive designs. According to the Oxford Centre for Evidence-Based Medicine classification [13], most included studies were Level IV evidence (41 case series and 16 case reports), followed by Level III studies (16 cohort studies and 1 case-control study), with only one Level II study (1 non-randomised clinical trial) and one Level I randomised control trial. The studies were published between 1996 and 2024 and originated from 17 different countries. The main characteristics of the included studies are summarised in Tables 1 and 2.

3.3. Characteristics of RCTs and observational studies

A total of 3234 patients undergoing vertebral augmentation procedures for metastatic thoracolumbar spinal fractures were identified, corresponding to a minimum of 5737 vertebral levels treated, as Shi X et al. do not report the total number of levels treated [14]. Patient characteristics were heterogeneous across studies. The weighted mean age was 64,1 years, with reported study means ranging from 53,5 to 78,9 years. Overall, 49.3% of the patients were male. Follow-up duration was reported in 37 studies, with a weighted mean follow-up of 12.3 months (median 12 months, range 1–118 months). Tumour type was reported in 54 studies and 16 included patients with a single tumour

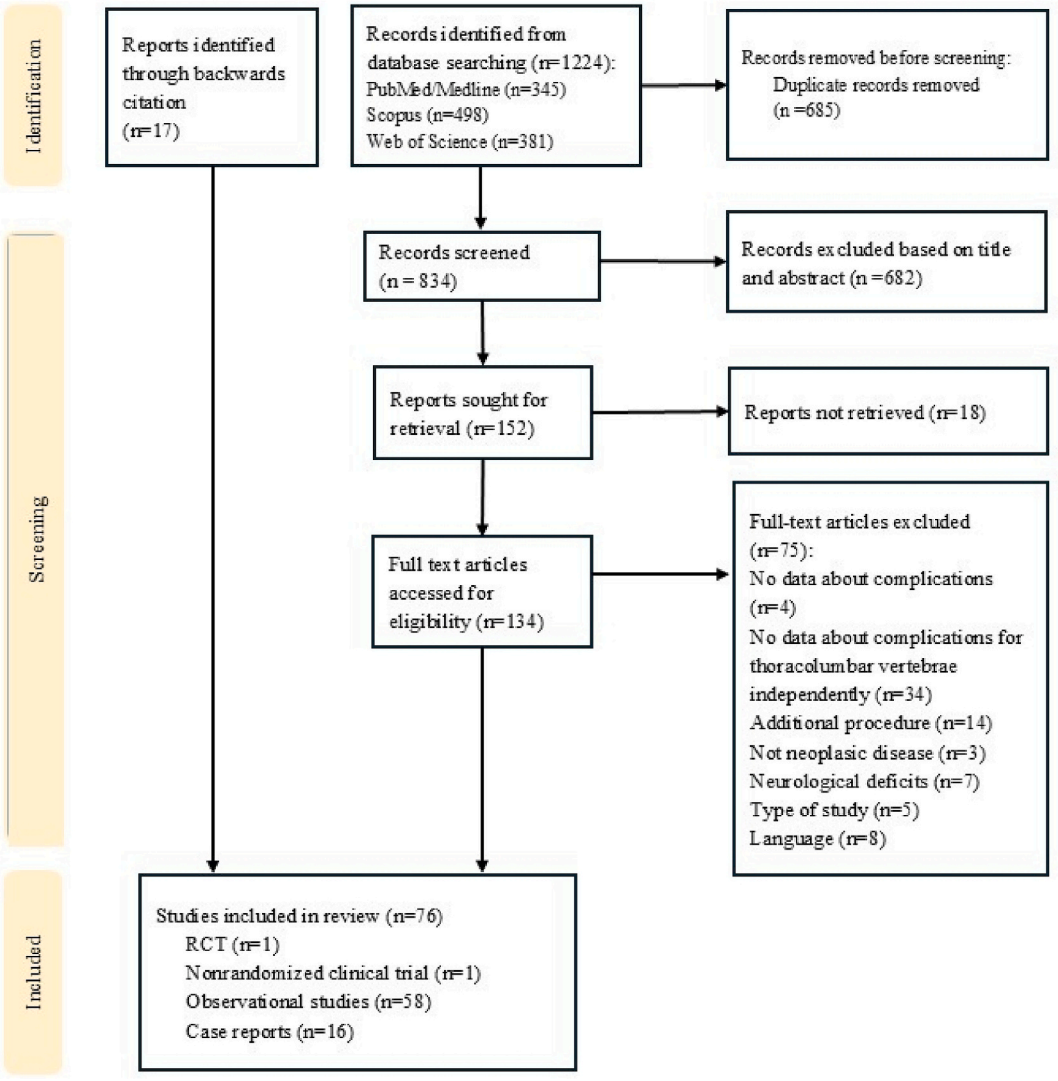


Fig. 1. Flowchart of study selection.

Table 1
Characteristics of included observational studies.

Study	Study type	Sample size	Levels treated & mean level per surgery	Mean age & gender	Primary tumour types		Procedure	Follow-up
1. Barr, JD et al. [15] 2000, USA	Cohort single arm	8 patients 13 vertebrae	Thoracolumbar spine - -	*no information*	Metastatic disease		Vertebroplasty *CT or fluoroscopy guided *With DSA	12 months (2–35)
2. Berenson, J et al. [57] 2011, Cancer Patient Fracture Evaluation (CAFE) study	RCT	68 patients 108 vertebrae	Thoracolumbar spine T5-L5 -	64,8 years 41% men	Multiple myeloma: 22 Breast: 16 Lung: 7	Prostate: 4 Other: 19	Kyphoplasty -	*no information*
3. Burton, AW et al. [49]2005, USA	Case series	3 patients 6 vertebrae	Thoracolumbar spine T12-L5 2 (1–3) levels/ session	64 years 33% men	Lung: 1 Breast: 1	Parathyroid: 1	Vertebroplasty Fluoroscopy guided	Up to 3 months or death
4. Caudana, R et al. [16] 2008, Italy	Cohort single arm	39 patients 62 vertebrae	Thoracic (56,4%) and lumbar (43,6%) spine - -	62 years 36% men	Breast: 17 Lung: 7 Kidney: 4 Multiple myeloma: 4 Colon: 3	Prostate: 1 Oesophagus: 1 Liver: 1 Neuroendocrine tumours: 1	Vertebroplasty CT guided	3,39 months (0,3-11,5)
5. Chen, G et al. [22] 2013, China	Case series	6 patients 6 vertebrae	Thoracic (50%) and lumbar (50%) spine T9-L5 1 level/ session	59,3 years 50% men	Breast: 2 Lung: 2	Liver: 1 Prostate: 1	Kyphoplasty Fluoroscopy guided	35 months (16–96)
6. Chen, L et al. [58] 2011, China	Case series	4 patients 4 vertebrae	Thoracic (25%) and lumbar (75%) spine T8-L4 -	57,75 years 75% men	Lung: 2 Prostate: 1	Pancreas: 1	Vertebroplasty Fluoroscopy guided	18 months (14–24)
7. Chen, KY et al. [40] 2009, Taiwan	Cohort single arm	31 patients 42 vertebrae	Thoracic (35,7%) and lumbar (64,2%) spine T3-L5 -	67 years 39% men	Lung: 10 Breast: 7 Gastrointestinal tract: 4 Hepatobiliary: 3 RCC: 1	Cervical: 1 Multiple myeloma: 1 Pancreatic: 1 Prostate: 1 Unknown: 1 Thyroid: 1	Vertebroplasty -	*no information*
8. Chew, C et al. [59] 2011, Scotland, UK	Cohort single arm	128 patients 264 vertebrae	Thoracolumbar spine - Up to 4 levels/ session	60 years 53% men	Multiple myeloma: 41 Breast: 22 Lung: 16	Lymphoma: 11 Renal: 8 Prostate: 5 Others: 25	Vertebroplasty Fluoroscopy guided	*no information*
9. Cui, Y et al. [52] 2022, China	Comparative retrospective cohort	230 patients 530 vertebrae	Thoracic (50,6%) and lumbar spine (49,4%) - -	66 years 57% men	Lung: 172 Breast: 82	Prostate: 78 Other: 198	Vertebroplasty Fluoroscopy guided	*no information*
10. Da Silva Herrero, CFP et al [60] 2014, Brazil	Cohort single arm	22 patients 34 vertebrae	Thoracic (30%) and lumbar (70%) spine T7-L5 -	56 years 36% men	Multiple Myeloma: 8 Breast: 6 Plasmacytoma: 2 Paraganglioma: 1 Oropharynx: 1	Fibro histiocytoma: 1 Cervix: 1 Multiple angiomatosis: 1 Gastric: 1	Kyphoplasty Fluoroscopy guided	8,5 months
11. Diamond, TH et al. [41] 2004, Australia	Case series	7 patients 14 vertebrae	Thoracolumbar spine T5-L1 Up to 7 levels/ session	69 years 43% men	Multiple Myeloma: 7		Vertebroplasty Fluoroscopy guided	6 weeks
12. Dudeney, BS et al [42] 2002, USA	Cohort single arm	18 patients 55 vertebrae	Thoracolumbar spine T6-L5 -	63,5 years -	Multiple Myeloma: 18		Kyphoplasty Fluoroscopy guided	*no information*
13. Eleraky, M et al [61] 2011, Germany	Cohort single arm	14 patients 30 vertebrae	Thoracic spine T1-T5 1–3 levels/session	62 years 57% men	Multiple Myeloma: 6 Lung: 3 Breast: 2	Melanoma: 1 Leiomiosarcoma: 1 Bladder: 1	Kyphoplasty Fluoroscopy guided	18 months (minimum 12 months)
14. Farrokhi, M et al. [23] 2012, Iran	Cohort single arm	25 patients 25 vertebrae	Thoracic (28%) and lumbar (72%) spine T3-L4 -	53,5 years 44% men	Breast: 14 Lung: 9	Prostate: 2	Vertebroplasty Fluoroscopy guided	*no information*
15. Fourney, DR et al. [38] 2003, USA	Cohort single arm	56 patients 97 vertebrae	Thoracolumbar spine T3-L5 1,7 (1–5) levels/ session	64 years (median) 55% men	Multiple Myeloma: 21 Breast: 9 Lung: 6 Lymphoma: 6	Prostate: 5 Renal cell: 2 Gastric: 2 Other: 5	Kyphoplasty and Vertebroplasty Fluoroscopy guided	Median 4,5 months

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Table 1 (*continued*)

Study	Study type	Sample size	Levels treated & mean level per surgery	Mean age & gender	Primary tumour types		Procedure	Follow-up
16. García-Maroto, R et al. [62] 2015, Spain	Cohort single arm	75 patients 122 vertebrae	Thoracic (36,9%) and lumbar (63,1%) spine - -	68 years 33,3% men	Multiple Myeloma: 30 Breast: 14 Lung: 8	Colon: 8 Prostate: 3 Others: 12	Kyphoplasty Fluoroscopy guided	11 months (3–36)
17. Garland, P et al. [43] 2011, England, UK	Cohort single arm	26 patients 57 vertebrae	Thoracolumbar spine T6-L5 1–3 levels/ session	59 years -	Multiple Myeloma: 26		Vertebroplasty Fluoroscopy guided	19 months
18. Guzik, G [44] 2017, Poland	Cohort single arm	73 patients 112 vertebrae	Thoracolumbar spine - Up to 3 levels/ session	71 years 36% men	Multiple Myeloma: 73		Vertebroplasty Fluoroscopy guided	*no information*
19. Ha KY et al [24]2015, Korea	Comparative cohort	27 patients 48 vertebrae	Thoracolumbar spine - -	66,6 years 41% men	Metastatic disease		Kyphoplasty and Vertebroplasty Fluoroscopy guided	16,8 months (6–33)
20. He, S et al. [63] 2020, China	Comparative cohort	147 patients 190 vertebrae	Thoracolumbar spine T6-L5 -	70 years 35% men	Metastatic disease		Kyphoplasty Fluoroscopy guided	12 months
21. Huber FX et al. [25] 2009, Germany	Cohort single arm	76 patients 190 vertebrae	Thoracolumbar spine T6-L5 2,5 levels/ session Up of 5 levels/ session	62 years (median) 59% men	Multiple Myeloma: 76		Kyphoplasty Fluoroscopy guided	*no information*
22. Jang JS et al. [45]2002, South Korea	Case series	27 patients 72 vertebrae	Thoracolumbar spine - -	- 67% men	Multiple Myeloma: 27		Vertebroplasty Fluoroscopy guided	*no information*
23. Julka, A et al. [46] 2014, USA	Cohort single arm	32 patients 76 vertebrae	Thoracic (45%) and lumbar (55%) spine - -	64,3 years 56% men	Multiple Myeloma: 32		Kyphoplasty Fluoroscopy guided	24 months
24. Kirkegaard, AO et al. [64] 2018, Denmark	Cohort single arm	30 patients 81 vertebrae	Thoracic (45,7%) and lumbar (54,3%) spine - Single level/session in 20% of patients and multiple levels/ session in 80% of patients	69,1 years 67% men	Multiple Myeloma: 13 Lung: 6 Chronic lymphocytic leukaemia: 5 Lymphoma: 2	Gastric: 1 Prostate: 1 Colon: 1 Ovarian: 1	Vertebroplasty Fluoroscopy guided	*no information*
25. Kırcelli et al. [65] 2018, Turkey	Cohort single arm	72 patients 82 vertebrae	Thoracolumbar spine - (1–3) levels/ session	78,93 years 24% men	Metastatic disease		Kyphoplasty Fluoroscopy guided	18,91 months
26. Kobayashi, T et al. [17] 2009, Japan	Prospective multi-institutional single-arm Phase I/II clinical study	33 patients 42 vertebrae	Thoracic (42,9%) and lumbar (57,1%) spine - 1–3 levels/ session	62 years 48% men	Lung: 7 Breast: 7 Colorectal: 7 Liver: 4 Multiple Myeloma: 3	Pancreatic: 2 Tongue: 1 Oesophageal: 1 Skin: 1	Vertebroplasty Fluoroscopy and CT guided	*no information*
27. Kodama, H et al. [18] 2007, Japan	Cohort single arm	6 patients 8 vertebrae	Thoracic (50%) and lumbar (50%) spine T1-L1 1–2 levels/ session	65,2 years 100% men	Hepatocellular: 6		Vertebroplasty CT guided	*no information*
28. König MA et al. [39] 2012, UK	Case series	11 patients 23 vertebrae	Thoracolumbar spine T2-L3 -	65 years 36% men	Breast: 3 Non-Hodgkin lymphoma: 2 Gastric: 1 Cervix:1 Prostate: 1	Lung: 1 Bladder: 1 Multiple Myeloma: 1	Kyphoplasty Fluoroscopy guided	74,4 weeks
29. Lee, B et al [19] 2009, England, UK	Cohort single arm	18 patients 33 vertebrae	Thoracolumbar spine - 1–3 levels/ session	69,3 years 39% men	Breast: 7 Prostate: 4 Renal: 2 Lung: 1	Oesophageal:1 Adenoid cystic carcinoma: 1 Cervix: 1 Other: 1	Vertebroplasty Fluoroscopy and CT guided	12 months (3–38)
30. 1 Li, Z et al. [11] 2014, China	Comparative cohort 1° (KP)	42 patients 83 vertebrae	Thoracolumbar spine - 1–4 levels/ session	68,4 years 43% men	Breast: 16 Lung: 9 Prostate: 6	Stomach: 6 Cervix: 3 Rectal: 2	Kyphoplasty Fluoroscopy guided	1 year
30.2 Li, Z et al. [11] 2014, China	Comparative cohort 2° (VP)	38 patients 76 vertebrae	Thoracolumbar spine - 1–4 levels/ session	65,2 years 50% men	Breast: 14 Lung: 9 Prostate: 8	Stomach: 5 Ovarian: 2	Vertebroplasty Fluoroscopy guided	1 year

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Table 1 (*continued*)

Study	Study type	Sample size	Levels treated & mean level per surgery	Mean age & gender	Primary tumour types		Procedure	Follow-up
31. Lim, B et al. [66] 2009, South Korea	Comparative cohort	102 patients 185 vertebrae	Thoracic (21%) and lumbar (79%) spine T6-L5 -	55 years 59% men	Breast: 24 Multiple Myeloma: 19 Lung: 16 Liver: 10 Rectal: 6	Unknown primary: 5 Stomach: 4 Colon: 4 Cervix: 2 Other: 12	Vertebroplasty Fluoroscopy guided	12,2 months
32. Liu, H et al. [67] 2017, China	Comparative cohort	234 patients 288 vertebrae	Thoracolumbar spine - -	63,7 years 74% men	Lung: 100 Breast: 97 Prostate: 14 Stomach: 11 Rectal: 8	Liver: 5 Renal: 5 Cervix: 2 Ovarian: 1	Kyphoplasty Fluoroscopy guided	12 months
33. Liu, Y et al. [20] 2023, China	Case-control	30 patients 30 vertebrae	Thoracic (20%) and lumbar (80%) spine - -	59,5 years 67% men	Lung: 30		Vertebroplasty CT guided	*no information*
34. Masala, S et al. [47] 2008, Italy	Cohort single arm	64 patients 198 vertebrae	Thoracic (47%) and lumbar (53%) spine - 3,09 levels/ session	71,4 years (median) 53% men	Multiple Myeloma: 64		Vertebroplasty Fluoroscopy guided	6 months
35. Moulin, B et al. [26] 2020, Japan	Cohort single arm	50 patients 397 vertebrae	Thoracic (60%) and lumbar (40%) spine - 7,9 (6–13) levels/ session	66 years 27% men	Multiple Myeloma: 13 Breast: 12 Prostate: 6	Lung: 6 Other: 13	Vertebroplasty Fluoroscopy guided	401 days
36. Nas, OF et al. [68] 2015, Turkey	Cohort single arm	52 patients 95 vertebrae	Thoracic (43,2%) and lumbar (56,8%) spine - 1–5 levels/ session	63,98 years 46% men	Breast: 14 Lung: 9 Prostate: 7 Renal: 5	Stomach: 5 Others: 12	Vertebroplasty Fluoroscopy guided	3 months
37. Pflugmacher, R et al [48] 2005, Germany	Cohort single arm	20 patients 48 vertebrae	Thoracolumbar spine T6-L5 -	62,4 years 100% men	Multiple Myeloma: 20		Kyphoplasty Fluoroscopy guided	Minimum of 1 year
38. Pflugmacher, R et al 2007, [69] Germany	Cohort single arm	31 patients 64 vertebrae	Thoracolumbar spine T6-L5 -	64,5 years 42% men	Prostate: 7 Cervix: 5 Rectum: 5 Ovary: 5	Lung: 3 Pancreas: 2 Breast: 2 Stomach: 2	Kyphoplasty Fluoroscopy guided	1 year in 28 patients
39. Pflugmacher, R et al 2008, [70] Germany	Cohort single arm	65 patients 99 vertebrae	Thoracolumbar spine - 1–4 levels/ session	66 years 57% men	Metastatic disease		Kyphoplasty Fluoroscopy guided	2 years in 41 patients
40. Qian, Z et al. [27]2011, China	Cohort single arm	48 patients 97 vertebrae	Thoracic (64%) and lumbar (36%) spine T5-L5 2,6 levels/ session	68,5 years 40% men	Breast: 16 Lung: 10 Prostate: 6 Stomach: 6 Cervical: 4	Rectal: 3 Ovarian: 3	Kyphoplasty Fluoroscopy guided	2 years in 39 patients
41. Rajah, G et al. [71] 2015, USA	Comparative cohort	45 patients 95 vertebrae in study group	Thoracic (55%) and lumbar (45%) spine - -	38% men	Multiple Myeloma: 21 Lung: 9 Breast: 6 Polycythaemia: 2 Colon: 1 Neck adenocarcinoma: 1	Prostate: 1 Renal: 1 Non-Hodgkin lymphoma: 1 Chronic myeloid leukaemia: 1 Lymphoma: 1	Kyphoplasty Fluoroscopy guided	*no information*
42. Ramos, L et al [28] 2005, Spain	Case series	12 patients 19 vertebrae	Thoracic (47%) and lumbar (53%) spine T9-L4 1–3 levels/ session	66 years 75% men	Multiple Myeloma: 12		Vertebroplasty Fluoroscopy guided	3,2 years (2–56 months)
43. Reyad, RM et al. [29] 2017, Egypt	Cohort single arm	63 patients 92 vertebrae	Thoracic (39%) and lumbar (61%) spine T8-L5 1–4 levels/ session	58 years 48% men	Breast: 15 Lung: 11 Hepatocellular: 11 Prostate: 9 Multiple Myeloma: 5	Thyroid: 5 Lymphoma: 3 Colon: 3 Rectal: 1	Vertebroplasty Fluoroscopy guided	12 weeks
44. Saracen, A et al [72] 2016, Poland	Cohort single arm	90 patients 137 vertebrae	Thoracolumbar spine - 1–3 levels/ session	- 30% men	Prostate: 36 Multiple Myeloma: 32	Breast: 20 Lung: 2	Vertebroplasty Fluoroscopy guided	*no information*
45. Shi X et al. [14] 2021, China	Comparative cohort	309 patients 309 vertebrae	Thoracolumbar spine - -	- 59% men	Metastatic disease		Vertebroplasty -	*no information*
46. Shimony, JS et al. [73] 2024, USA	Comparative cohort	50 patients 129 vertebrae	Thoracolumbar spine T1-L5	66,7 years 50% men	Multiple Myeloma: 14 Lung: 13 Breast: 8	Adenocarcinoma: 1 Pancreas: 1	Vertebroplasty Fluoroscopy guided	23,8 weeks

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Table 1 (*continued*)

Study	Study type	Sample size	Levels treated & mean level per surgery	Mean age & gender	Primary tumour types		Procedure	Follow-up
			2,6 (1–7) levels/ session		Prostate: 4 Lymphoma: 3 Unknown primary: 3	Neuroendocrine: 1 Bladder: 1 Uterine: 1		
47. Tekin, SB et al. [74] 2020, Turkey	Cohort single arm	42 patients 76 vertebrae	Thoracolumbar spine T8-L5 -	64,2 years 36% men	Multiple Myeloma: 42		Vertebroplasty Fluoroscopy guided	12,6 months
48. Tian, Q et al. [30] 2016, China	Cohort single arm	39 patients 51 vertebrae	Thoracolumbar spine - 1–3 levels/ session	60,1 years 62% men	Lung: 15 Prostate: 11 Breast: 9	Liver: 3 Colon: 1	Vertebroplasty Fluoroscopy guided	14,5 months
49. Trumm, CG et al. [31] 2008, Germany	Cohort single arm	53 patients 76 vertebrae	Thoracic (72%) and lumbar (28%) spine - 1–3 levels/ session	62 years 2% men	Breast: 53		Vertebroplasty Fluoroscopy guided	9,2 months
50. Vrionis, FR et al [75] 2005, USA	Cohort single arm	50 patients 128 vertebrae	Thoracic (65%) and lumbar (35%) spine T3-L5 2,56 (1–6) levels/ session	63 years 34% men	Multiple Myeloma: 23 Lung: 11 Breast: 8 Prostate: 4	Colon: 1 Oesophageal: 1 Uterine: 1 Sarcoma: 1	Kyphoplasty Fluoroscopy guided	9 months
51. Wang, H et al. [76] 2023, China	Comparative cohort	36 patients 36 vertebrae	Thoracic (39%) and lumbar (61%) spine T5-L5 -	61,1 years 44% men	Breast: 11 Lung: 7 Prostate: 6 Stomach: 3 Liver: 2	Kidney: 2 Thyroid: 1 Colon: 1 Oesophagus: 1 Nasopharynx: 1 Cervical: 1	Kyphoplasty -	Up to 24 months
52. Wang, L et al. [32] 2022, China	Comparative retrospective cohort	64 patients 113 vertebrae	Thoracic (42%) and lumbar (58%) spine - -	65 years 56% men	Metastatic disease		Vertebroplasty Fluoroscopy guided	3 months
53. Wang, Y et al. [33] 2016, China	Comparative cohort	41 patients 73 vertebrae	Thoracolumbar spine - -	56,2 years 44% men	Breast: 13 Lung: 9 Prostate: 6 Stomach: 5 Liver: 4	Rectal: 3 Cervical: 2 Renal: 2 Ovarian: 1	Kyphoplasty Fluoroscopy guided	1 year
54. Weill, A et al. [34] 1996, France	Case series	21 patients 27 vertebrae	Thoracic (44,4%) and lumbar (55,6%) spine T6-L4 1–3 levels/ session	57,1 years 48% men	Breast: 8 Lung: 3 Kidney: 3 Pheochromocytoma: 2	Prostate: 2 Uterus: 1 Oesophagus: 1 Bladder: 1	Vertebroplasty Fluoroscopy guided	7,1 months
55. Wu, W et al. [35] 2022, China	Cohort single arm	31 patients 58 vertebrae	Thoracic (32,8%) and lumbar (67,2%) spine - 1–6 levels/ session	58,5 years 39% men	Lung: 11 Breast: 7 Kidney: 4 Prostate: 2 Rectal: 1	Oesophagus: 1 Stomach: 1 Colon: 1 Ovarian: 1	Kyphoplasty Fluoroscopy guided	1 year in 27 patients
56. Wu, W et al. [36]2022, China	Comparative cohort	59 patients 59 vertebrae	Thoracic (36%) and lumbar (64%) spine - 1 level/ session	60,07 years 59% men	Metastatic disease		Kyphoplasty Fluoroscopy guided	30,33 months
57. Yamada, K et al. [21] 2004, Japan	Case series	4 patients 6 vertebrae	Thoracolumbar spine T12-L5 1–2 levels/ session	58,8 years 75% men	Mandibular: 1 Gastric: 1	Liver: 1 Colon: 1	Vertebroplasty CT guided	10–18 weeks
58. Yan, B C et al. [37] 2022, China	Comparative cohort	87 patients 111 vertebrae	Thoracolumbar spine T1-L5 -	58 years 48,3% men	Lung: 42 Liver: 17 Breast: 13 Thyroid: 10 Kidney: 7 Others: 7	Colon: 6 Multiple Myeloma: 3 Bladder: 2 Oesophageal: 2 Haemangioma: 2	Vertebroplasty Fluoroscopy guided	Up to 12 months
59. Zhang, H et al. [77] 2017, China	Case series	13 patients 13 vertebrae	Thoracolumbar spine - -	58,3 years 31% men	Breast: 4 Lung: 4 Prostate: 2	Bowel: 1 Kidney: 1 Liver: 1	Kyphoplasty Fluoroscopy guided	*no information*
60. Zhang, J et al. [50] 2023, China	Comparative cohort	28 patients 37 vertebrae	Thoracic (49%) and lumbar (51%) spine T7-L5 -	64,96 years 54% men	Lung: 28		Kyphoplasty Fluoroscopy guided	*no information*

type. Across the overall sample, the most frequent primary tumours were multiple myeloma (28.0%), breast cancer (21.6%), lung cancer (19.8%), and prostate cancer (8.0%).

Most of the studies performed percutaneous vertebroplasty, while in 23 the procedure was balloon kyphoplasty, and in three studies both techniques were used. The procedures were guided by fluoroscopy in most cases (82,5% of included studies), with a smaller number guided by CT scan [15–21]. Twenty-two studies mentioned performing routine CT evaluation in the post-op period [11,16–18,21–37], and one study if there was cement leakage visualised intra-operatively [38]. Overall, 44,3% of the included studies had a CT evaluation of the spine, either intra-operatively or post-operatively.

When reported, the mean injected cement volume ranged from 1.5 to 5.87 mL per treated vertebra, although higher volumes were occasionally described in lumbar vertebrae [37,39]. Information regarding the number of vertebral levels treated per procedure was reported in 39 studies. Among these, the majority (51.3%) performed vertebral augmentation in a maximum of 3 levels per session

Table 2
Characteristics of included case-reports.

Study	Year	Performed procedure	Complication	Time until symptoms	Outcome
1. Amoretti, N et al. [78]	2007	Vertebroplasty	**Cement migration to the aorta** Prevertebral cement leakage during the procedure (intra-operative CT demonstrated cement migration through the lumbar artery into the abdominal aorta)	Intra-operative	Asymptomatic
2. Chen, YJ et al. [55]	2007	Vertebroplasty	**Needle tract tumour seeding** MRI showed epidural and paraspinal metastasis along the trajectory of the needle	3 months	Persistent paraparesis despite decompression surgery
3. Chevallier, M et al. [79]	2021	Vertebroplasty	**Cement migration to cardiac chambers** Intra-operative paravertebral venous cement leakage (CT demonstrated a cement filament extending into the right heart chambers with pulmonary emboli)	1 month	Failed endovascular removal Subsequent pulmonary embolism and death
4. Cosar M et al. [56]	2009	Vertebroplasty	**Epidural cement leakage** Cement extravasation into the epidural space, causing spinal cord compression	Post-operative	Decompressive laminectomy Persistent paraplegia
5. Cruz, JP et al. [80]	2014	Kyphoplasty	**Tumour progression** MRI showed tumour extension along the cannula trajectory	2,5 weeks	Treated with palliative radiotherapy
6. Esmende, SM et al. [81]	2013	Kyphoplasty	**Tumour retropulsion with spinal canal compression** CT/MRI showed a tumour mass displaced into the ventral spinal canal	Post-operative	Emergent decompression surgery with full recovery
7. Gorospe, L et al. [82]	2018	Vertebroplasty	**Needle tract tumour seeding** Subcutaneous metastatic nodule along the needle path	6 months	Asymptomatic lesion
8. Lee, CH et al. [83]	2014	Vertebroplasty	Needle tract tumour seeding Skin metastasis and epidural tumour extension along the needle tract	5 months	Surgical excision and spinal fusion Death due to disease progression
9. Li, Z et al. [11]	2013	Vertebroplasty	**Cement leakage into the inferior vena cava** Venography showed communication between vertebral veins and IVC, leading to cement migration	Intra-operative	Asymptomatic IVC filter captured cement
10. Mao, HQ et al. [84]	2011	Vertebroplasty	**Extradural aracnoyd cyst** Radiculopathy caused by cyst formation related to dural injury	5 months	Surgical decompression with full recovery
11. Mercer, J et al. [85]	2020	Vertebroplasty	**Pulmonary cement embolism** CT demonstrated multiple cement emboli in pulmonary arteries	2 months	Asymptomatic
12. Morghen, I et al. [86]	2009	Vertebroplasty	**Tumour extrusion into the spinal canal** MRI showed extrusion of tumour tissue into spinal canal without neurological deficit	24 h	Radiotherapy Pain resolution
13. Ratliff J et al. [87]	2001	Vertebroplasty	**Cement leakage into the spinal canal** Cement extravasation through neural foramina, causing neurological deficit	Intra-operative	Surgical cement removal with full recovery
14. Sabuncuoğlu, H et al. [88]	2008	Vertebroplasty	**Intradural cement leakage** Incorrect cannula trajectory causing dural injury and intradural cement deposition	Intra-operative	Laminectomy and cement removal with neurological improvement
15. Shi, G et al. [89]	2020	Vertebroplasty	**Transient paraparesis** Likely due to anaesthetic infiltration into the spinal canal rather than cement leakage	Post-operative	Complete resolution after 4 h
16. Tropeano, MP et al. [90]	2017	Vertebroplasty	**Subdural haematoma** Iatrogenic dural injury	3 weeks	Surgical evacuation with full recovery

[15–17,19,23–25,27–29,38,40–48].

Substantial clinical and methodological heterogeneity was observed across studies, including differences in study designs, populations, procedural techniques, and complication detection methods. Although all studies involved metastatic spinal disease, some included only pathological fractures, while others considered vertebral metastases alone as inclusion criteria. Although none included patients with neurological deficits, some studies included vertebrae with epidural tumour involvement, while others excluded posterior wall destruction or epidural extension. Methodological heterogeneity was also substantial due to variability in study design. In addition, the criteria used to define complications varied between studies. While some authors considered only symptomatic events as complications, others reported all adverse events, including asymptomatic findings detected on imaging. Some studies documented a broad range of medical events during follow-up, whereas others reported only complications directly attributable to vertebral augmentation. Among studies that considered cement leakage as a complication, some reported all types of leakage, whereas others excluded specific patterns such as leakage into paravertebral soft tissues. Moreover, while approximately half of the included studies systematically performed postoperative CT to detect cement leakage, the remaining studies relied primarily on intraoperative fluoroscopy or postoperative radiographs, which may underestimate the true incidence of leakage. Substantial heterogeneity was also observed in the reported complication rates across studies. Given the variability in study populations, methodologies, definitions of complications, and diagnostic

approaches, a quantitative meta-analysis was deemed inappropriate. Therefore, the results were synthesised descriptively.

3.4. Complications of vertebral augmentation procedures

Complications of vertebral augmentation procedures are summarised in Table 3.

The mean reported complication rate was 24.3% across studies, corresponding to a weighted mean of 25.5% when adjusted for study sample size. Six studies reported no complications [18,40,41,47,49,50] Fig. 2. Of these, the study population was under 10 patients in three [18,41,49] and ranged from 28 to 61 patients in the other three. [40,47,50]

The most frequently described complication was cement leakage, reported in 50 articles (82%), with rates ranging from 1,3% to 98.7%. The mean leakage rate across studies was 19.3%, with a weighted mean of 20.7% when adjusted for study sample size. Studies using CT either for procedural guidance or postoperative evaluation reported significantly higher leakage rates compared with studies without CT (28.6% vs 11.9%, $p = 0.004$). No significant association was found between the type of vertebral augmentation procedure and the reported cement leakage rate ($p = 0.27$). Similarly, the number of vertebral levels treated per session was not significantly associated with cement leakage frequency ($p = 0.45$).

Of the 50 studies reporting cement leakage, only 37 described the pattern of leakage. The most common was venous leakage (mean 27.3%), followed by intradiscal leakage (26.8%), soft tissue leakage (22.0%), intracanal leakage (14.9%) and other types of leakage, such as around the injection site (5.1%). Some studies did not report all types of leaks, which could affect the understanding of the true pattern of cement leakage.

Other complications were rare across the included studies. The most frequently reported events were postprocedural pain (mean 1.8%), pulmonary cement embolism (mean 1.1%), and adjacent level fracture (mean 0.8%), while all other complications occurred in less than 0.5% of cases, including local haematoma (0.4%), fever (0.3%), pulmonary embolism or deep vein thrombosis (0.2%), inferior vena cava embolism (0.08%), procedural failure (0.09%), bleeding (0.05%), superficial infection (0.03%), and balloon asymptomatic rupture (0.02%).

39 of the 60 included studies reported no symptomatic complications. The mean rate of symptomatic complications across studies was 3,8%, with a weighted mean of 3,4% when adjusted for study sample size, ranging from 1% to 74,6%. The latter corresponds to the study by Reyad, RM et al., where 61,9% of patients presented with post-procedural pain, which may simply reflect the use of local anaesthesia. No significant association was observed between symptomatic complication rates and the type of vertebral augmentation procedure ($p = 0.274$) or the number of vertebral levels treated per session ($p = 0.375$). Studies using CT for intraoperative guidance or postoperative evaluation reported higher mean rates of symptomatic complications (5.1% vs 2.8%), although this difference was not statistically significant ($p = 0.918$). Among symptomatic complications, the most frequently reported were post-procedural pain (21.3%), fever (12.6%), local haematoma (7.8%), and deep vein thrombosis (6.8%). Less frequently, adjacent-level fracture (5.0%), pulmonary cement embolism (4.0%), and symptomatic cement leakage.

3.5. Complications described in case reports

Case reports identified several rare complications associated with vertebral augmentation procedures.

Case reports included descriptions of cement embolism, symptomatic cement leakage with neurological deficits, needle-tract tumour seeding, tumour extrusion beyond the vertebral body, cement migration to cardiac structures, subdural haematoma, extradural cyst, and transient paraparesis.

Several of these complications, particularly symptomatic cement leakage and embolic events, were also reported in larger observational studies included in this review. In contrast, complications such as needle tract tumour seeding, tumour extrusion, and cement migration to cardiac structures were described almost exclusively in case reports.

3.6. Quality assessment

The methodological quality of the included analytical studies was assessed using the JBI critical appraisal tools according to study design (see Appendix B). Overall, the risk of bias among cohort and comparative studies was moderate, with the proportion of positive responses across JBI items ranging from approximately 44% to 100%. Consequently, the findings of the present review should be interpreted in the context of the observational nature of the available evidence.

4. Discussion

This systematic review aimed to synthesise complications associated with vertebral augmentation in metastatic spinal disease.

Overall, vertebroplasty and kyphoplasty appear safe for pain relief and spinal stabilisation in oncologic patients. After adjustment for sample size, the overall complication rate was 25.5%, while symptomatic complications occurred in only 3,4%.

4.1. Cement leakage

Several previous systematic reviews have attempted to identify risk factors for complications after vertebral augmentation. A systematic review published in 2024 suggested that posterior cortical disruption, vertebral body fracture, and the number of treated vertebrae may increase the risk of cement leakage. However, that review included both percutaneous vertebral augmentation and open surgical techniques, which may limit direct comparison with the present review [51]. Similarly, previous systematic reviews evaluating vertebral augmentation for malignant vertebral compression fractures have reported cement leakage as the most frequently observed complication, although symptomatic events remain uncommon [7,52].

Patients with metastatic spinal disease often exhibit extensive vertebral destruction, including cortical posterior wall involvement. Coupled with the rich vascular supply of the vertebral column, this may increase the risk of cement leakage. Reduced bone quality related to age and systemic anticancer treatments further increases their risk and susceptibility to complications [53] [54]. In our review, cement leakage was the most frequently reported complication, consistent with prior literature. However, its clinical impact appears limited, as most cases were asymptomatic (20.7% of overall complications vs 3,2% of symptomatic ones). This finding suggests that radiological cement leakage should not automatically be interpreted as a clinically relevant adverse event. In some studies, small asymptomatic cement leaks were not considered complications and were therefore excluded from complication rates, increasing uncertainty regarding the true clinical relevance of cement leakage. The heterogeneity among the included studies in the definition of included patients did not allow for a statistical evaluation of the influence of posterior vertebral wall compromise in the occurrence of leakage.

Cement leakage can occur through intradiscal, intravascular, or paravertebral pathways, and was associated with complications such as nerve compression, pulmonary and intracardiac embolism, adjacent vertebral fractures, and local pain, which is consistent with previous reports in the literature [53]. Neurological complications are rare but particularly serious, as described by Chen, YJ et al. and Cosar M et al. [55,56]. However, it remains unclear how many patients develop symptoms after cement leakage, and the factors that increase the risk of clinically significant complications are not fully understood. Future studies should distinguish between radiographic and clinically

Table 3
Treatment characteristics & reported complications(*When available, complication rates were extracted according to the denominator reported in each study (per vertebra or per patient). In studies where this was not specified, cement leakage rates were calculated relative to the number of treated vertebrae, whereas other complications were calculated relative to the total number of procedures.)*

Study	Procedure details	Imaging details	Cement details	Complications	Symptomatic complications	Type of cement leakage
1. Barr, JD et al.	Vertebroplasty * 8 patients * 13 vertebrae	Fluoroscopy or CT * CT post-op	2–3 mL / thoracic vertebra 3–5 mL / lumbar vertebrae	**Total: 7,7%** * Cement Leakage: 7.7% (1/13)	**0%**	Venous system: 100% (1/1)
2. Berenson, J et al.	Kyphoplasty * 68 patients	- * No post-op CT	*No information*	**Total: 12,5%** * Cement Leakage: 4,7% (3/68) * Asymptomatic balloon rupture: 1,5% (1/68) * Procedure pain: 1,6% (1/68) * Superficial infection: 1,6% (1/68) * Adjacent level fracture: 3,1% (2/68)	**Total: 4,7%** * Superficial infection: 1,6% (1/68) * Adjacent level fracture: 3,1% (2/68) ** *one of the fractures occurred 1 day post-op, in a patient with intradiscal cement leakage*	*No information*
3. Burton, AW et al.	Vertebroplasty * 3 patients * 6 vertebrae	Fluoroscopy * No post-op CT	3–4 mL / vertebra	**0%**	**0%**	–
4. Caudana, R et al.	Vertebroplasty * 39 patients * 62 vertebrae	CT- guided * CT post-op	Mean 5.7 mL/ vertebra	**Total: 75,2%** * Cement Leakage: 72,5% (45/62) * Pulmonary Cement Embolism: 2,6% (1/39)	**Total: 5,9%** * Cement Leakage: 3,2% (2/62) ** *radiculopathy in leakage to the epidural space (medical treatment)* * Cement PE: 2,7% (1/39) ** *mild dyspnoea (medical treatment)*	Paravertebral soft tissues: 17,8% (8/45) Intracanal: 28,9 (13/45) Intradiscal: 2,2% (1/56) Venous system: 17,8% (8/45)
5. Chen, G et al.	Kyphoplasty * 6 patients * 6 vertebrae	Fluoroscopy * CT post-op	Mean 3.3 mL /vertebra (2–5)	**Total: 16,7%** * Cement Leakage: 16,7% (1/6)	**0%**	Venous system: 100% (1/1)
6. Chen, L et al.	Vertebroplasty * 4 patients * 4 vertebrae	Fluoroscopy * No post-op CT	Mean 2,9 mL/ vertebra (2,2-3,5)	**Total: 25%** * Cement Leakage: 25% (1/4)	**0%**	Venous system: 100% (1/1)
7. Chen, KY et al.	Vertebroplasty * 31 patients * 42 vertebrae	- * -	–	**0%**	**0%**	–
8. Chew, C et al.	Vertebroplasty * 128 patients * 158 procedures * 264 vertebrae	Fluoroscopy * -	< 5 mL / vertebra	**Total: 3,7%** * IVC cement embolism: 0,6% (1/158) * Pulmonary cement embolism: 1,9% (3/158) * Local haematoma: 0,6% (1/158) * Cement leakage: 0,6% (1/158) ** *"Asymptomatic small para-vertebral leaks were not considered complications or accounted for"*	**Total: 1,2%** Local haematoma: 0,6% (1/158) Cement leakage: 0,8% (1/158) * transient loss of sensation in T1 dermatome	Intracanal: 100% (1/1)
9. Cui, Y et al.	Vertebroplasty * 230 patients * 530 vertebrae	Fluoroscopy * No post-op CT	–	**Total: 34,9%** * Cement Leakage: 34,9% (185/530)	**0%**	Intradiscal or paravertebral: 47,5% (97/204) Intracanal: 18,2% (37/204) Intravenous: 34,3% (70/204)
10. Da Silva Herrero, CFP et al	Kyphoplasty * 22 patients * 34 vertebrae	Fluoroscopy * -	–	**Total: 20,8%** * Cement Leakage: 11,8% (4/34) * DVT with PE: 4.5% (1/22) *Adjacent level fracture: 4.5% (1/22)	**Total: 9%** * DVT with PE: 4.5% (1/22) *Adjacent level fracture: 4.5% (1/22)	Paravertebral soft tissues: 50% (2/4) Intracanal: 25% (1/4) Intradiscal: 25% (1/4)
11. Diamond, TH et al.	Vertebroplasty * 7 patients * 14 vertebrae	Fluoroscopy * No post-op CT	1–6 mL / vertebra	**0%**	**0%**	–
12. Dudeney, BS et al	Kyphoplasty * 18 patients * 27 procedures * 55 vertebrae	Fluoroscopic * No post-op CT	–	**Total: 4%** * Cement Leakage: 4% (2/55)	**0%**	Paravertebral soft tissues: 50% (1/2) Intracanal: 50% (1/2)

(*continued on next page*)

Table 3 (*continued*)

Study	Procedure details	Imaging details	Cement details	Complications	Symptomatic complications	Type of cement leakage
13. Eleraky, M et al	Kyphoplasty * 14 patients * 30 vertebrae	Fluoroscopy * -	3 mL / vertebra (1,5–4,5 mL)	**Total: 10%** * Cement Leakage: 10% (3/30)	**0%**	Intradiscal: 100%
14. Farrokhi, M et al.	Vertebroplasty * 25 patients * 25 vertebrae	Fluoroscopy * -	4,5 mL / vertebra (3–6 mL)	**Total: 24%** * Cement Leakage: 24% (6/25)	**0%**	Paravertebral soft tissues: 45,5% (5/11) Intracanal: 18,1% (2/11) Intradiscal: 36,4% (4/11)
15. Fourney, DR et al.	Kyphoplasty & Vertebroplasty * 56 patients * 97 vertebrae	Fluoroscopy * CT post-op if cement leakage was detected intra-op	2–8 mL / vertebra	**Total: 6,2%** * Cement Leakage: 6,2% (6/97)	**0%**	Paravertebral soft tissues: 16,7% (1/6) Intradiscal: 83,3% (5/6)
16. García-Maroto, R et al.	Kyphoplasty * 75 patients * 122 vertebrae	Fluoroscopy * -	–	**Total: 5,7%** * Cement Leakage: 5,7% (7/122)	**0%**	Paravertebral soft tissues: 28,6% (2/7) Intradiscal: 71,4% (5/7)
17. Garland, P et al.	Vertebroplasty * 26 patients * 57 vertebrae	Fluoroscopy * No post-op CT	–	**Total: 28,4%** * Cement Leakage: 5,3% (5/57) * Post-procedural pain: 23,1% (6/26)	**Total: 26,6%** * Cement Leakage: 3,5% (2/57) ** transient pain * Post-procedural pain: 23,1% (6/26) ** resolved within 24 h	–
18. Guzik, G	Vertebroplasty * 73 patients * 112 vertebrae	Fluoroscopy * -	–	**Total: 4,1%** * Cement Leakage: 2,7% (2/73) * Pulmonary cement embolism: 1,4% (1/73)	**0%**	Intracanal: 100% (2/2)
19. Ha KY et al	Vertebroplasty & Kyphoplasty * 27 patients * 48 vertebrae	Fluoroscopy * -	4,3 mL / vertebra	**Total: 20,8%** * Cement Leakage: 20,8% (10/48)	**0%**	Paravertebral soft tissues: 40% (1/10) Intracanal: 10% (1/10) Intradiscal: 50% (5/10)
20. He, S et al.	Kyphoplasty * 147 patients * 190 vertebrae	Fluoroscopy * No post-op CT	3,88 mL / vertebrae	Total: 6% * Cement Leakage: 2,6% (5/190) * Fever: 3,4% (5/147)	**0%**	Paravertebral soft tissue: 80% (4/5) Intradiscal: 20% (1/5)
21. Huber FX et al.	Kyphoplasty * 76 patients * 190 vertebrae	Fluoroscopy * CT post-op	–	**Total: 19,7%** * Cement Leakage: 18,4% (35/190) * Pulmonary cement embolism: 1,3% (1/76)	**Total: 1,3%** * Paraparesis resolved with surgery due to cement intracanal leakage in thoracic vertebra: 1,3% (1/73)	–
22. Jang JS et al.	Vertebroplasty * 27 patients * 72 vertebrae	Fluoroscopy * -	–	**Total: 11,1%** * Pulmonary Cement Embolism: 11,1% (3/27)	**Total: 7,4%** * Pulmonary cement embolism: 7,4% (2/27)	–
23. Julka, A et al.	Kyphoplasty * 32 patients * 76 vertebrae	Fluoroscopy * No post-op CT	3 mL / vertebra	**Total: 37,5%** * Cement Leakage: 37,5% (12/32)	**0%**	Paravertebral soft tissue: 16,7% (2/12) Intradiscal: 50% (6/12) Intravenous: 25% (3/12) Unknown: 8,3% (1/12)
24. Kirkegaard, AO et al.	Vertebroplasty * 30 patients * 81 vertebrae	Fluoroscopy * No post-op CT	–	**Total: 14,8%** * Cement Leakage: 14,8% (12/81)	**0%**	–
25. Kırcelli et al.	Kyphoplasty * 72 patients * 82 vertebrae	Fluoroscopy * No post-op CT	5,21 mL / vertebra	**Total: 23,5%** * Cement Leakage: 8,3% (6/72) *Adjacent level fracture: 15,2% (11/72)	**0%**	–
26. Kobayashi, T et al.	Vertebroplasty * 33 patients * 42 vertebrae	Fluoroscopy & CT-guided * CT post-op	3,5 mL / vertebra (1–8 mL)	**Total: 3%** * Bleeding at puncture site: 3% (1/33)	**Total: 3%** * Bleeding at puncture site: 3% (1/33) ** *stopped after 5 min compression*	–
27. Kodama, H et al.	Vertebroplasty * 6 patients * 8 vertebrae	CT guided * CT post-op	–	**0%**	**0%**	–
28. König MA et al.	Kyphoplasty * 11 patients * 23 vertebrae	Fluoroscopy * No post-op CT	5,1 mL / vertebra (1–12 mL)	**Total: 36,4%** * Cement Leakage: 18,2% (2/11) * DVT: 9,1% (1/11) * Adjacent level fracture: 9,1% (1/11)	**Total: 18,2%** * DVT: 9,1% (1/11) * Adjacent level fracture: 9,1% (1/11)	Intracanal: 100% (2/2)
29. Lee, B et al	Vertebroplasty * 18 patients * 33 vertebrae	Fluoroscopy & CT-guided * -	–	**Total: 5,6%** * Cement Leakage: 5,6% (1/18)	**Total: 5,6%** * Cement Leakage: 5,6% (1/18) ** transient pain (<24 h)	Puncture site: 100%

(*continued on next page*)

Table 3 (*continued*)

Study	Procedure details	Imaging details	Cement details	Complications	Symptomatic complications	Type of cement leakage
30.1 Li, Z et al.	Kyphoplasty * 42 patients * 83 vertebrae	Fluoroscopy * CT post-op	–	**Total: 16,9%** * Cement Leakage: 16,9% (14/83)	**0%**	Paravertebral soft tissue: 28,6% (4/14) Intradiscal: 7,1% (1/14) Intravenous: 64,3% (9/14)
30.2 Li, Z et al.	Vertebroplasty * 38 patients * 76 vertebrae	Fluoroscopy * CT post-op	–	**Total: 30,3%** * Cement Leakage: 30,3% (23/76)	**0%**	Paravertebral soft tissue: 30,4% (7/23) Intradiscal: 13,1% (3/23) Intravenous: 56,5% (13/23)
31. Lim, B et al.	Vertebroplasty * 102 patients * 185 vertebrae	Fluoroscopy * No post-op CT	3,62 mL / vertebra	**Total: 17,3%** * Cement Leakage: 17,3% (32/185)	**0%**	Intracanal: 40,1% (13/32) Intradiscal: 40,1% (13/32) Intravenous: 18,8% (6/32)
32. Liu, H et al.	Kyphoplasty * 234 patients * 288 vertebrae	Fluoroscopy * -	–	**Total: 6,8** * Cement Leakage: 6,4% (15/234) * Pulmonary cement embolism: 0,4% (1/234)	**0%**	Paravertebral soft tissue: 53,3% (8/15) Intradiscal: 46,7% (7/15)
33. Liu, Y et al.	Vertebroplasty * 30 patients * 30 vertebrae	CT-guided	3,47 mL / vertebra	**Total: 53,3%** * Cement Leakage: 40% (12/30) * Haematoma at puncture site: 13,3% (4/30)	**Total: 13,3%** * Haematoma at puncture site: 13,33%	–
34. Masala, S et al.	Vertebroplasty * 64 patients * 198 vertebrae	Fluoroscopy * No post-op CT	–	**0%**	**0%**	–
35. Moulin, B et al.	Vertebroplasty * 50 patients * 397 vertebrae	Fluoroscopy * CT post-op * Chest-CT performed in 38 patients	–	**Total: 57%** * Pulmonary cement embolism: 47% (18/38) * Post-operative pain: 4% (2/50) * Post-operative transient alteration of general status: 2% (1/50) * Retroperitoneal haematoma: 2% (1/50) * Lumbar artery puncture: 2% (1/50)	**Total: 10%** * Pulmonary cement embolism: 2% (1/50) * Post-operative pain: 4% (2/50) * Post-operative transient alteration of general status: 2% (1/50) * Lumbar artery puncture: 2% (1/50) ** *embolization intra-op*	–
36. Nas, OF et al.	Vertebroplasty * 52 patients * 95 vertebrae	Fluoroscopy * -	4,91 mL / vertebra (1,5–9 mL)	**Total: 45,3%** * Cement Leakage: 45,3% (43/95)	**0%**	Intradiscal: 46,5% (20/43) Intravenous: 46,5% (20/43) Intradiscal and intravenous: 7% (3/43)
37. Pflugmacher, R et al	Kyphoplasty * 20 patients * 48 vertebrae	Fluoroscopy * No post-op CT	4,6 mL / vertebra (2,6–7,9 mL)	**Total: 15,4%** * Cement Leakage: 10,4% (5/48) * Adjacent level fracture: 5% (1/20)	**Total: 5%** * Adjacent fracture: 5% ** *need for surgical intervention*	Paravertebral soft tissue: 40% (2/5) Intradiscal: 60% (3/5)
38. Pflugmacher, R et al	Kyphoplasty * 31 patients * 64 vertebrae	Fluoroscopy * No post-op CT	–	**Total: 12,5%** * Cement Leakage: 12,5% (8/64)	**0%**	Paravertebral soft tissues: 25% (2/8) Intradiscal: 50% (4/8) Intravenous: 25% (2/8)
39. Pflugmacher, R et al	Kyphoplasty * 65 patients * 99 vertebrae	Fluoroscopy * No post-op CT	–	**Total: 20,1%** * Cement Leakage: 12,1% (12/99) * Adjacent fracture: 8% (5/65)	**0%**	–
40. Qian, Z et al.	Kyphoplasty * 48 patients * 97 vertebrae in study group	Fluoroscopy * CT post-op	–	**Total: 18,6%** Cement Leakage: 18,6% (18/97)	**0%**	Paravertebral soft tissue: 22,2% (4/18) Intradiscal: 16,7% (3/18) Intravenous: 61,1% (11/18)
41. Rajah, G et al.	Kyphoplasty * 45 patients * 75 vertebrae	Fluoroscopy	3,6 mL	**Total: 18,6%** * Cement Leakage: 13,3% (10/75) * Failure (retropulsion of cement into canal): 5,3% (4/75)	**Total: 5,3%** * Failure: 5,3% ** need for surgical decompression and fusion	–
42. Ramos, L et al	Vertebroplasty * 12 patients * 19 vertebrae	Fluoroscopy * CT post-op	–	**Total: 84%** Cement Leakage: 84% (16/19)	**0%**	Intracanal: 12,5% (2/16) Intradiscal: 18,7% (3/

(*continued on next page*)

Table 3 (*continued*)

Study	Procedure details	Imaging details	Cement details	Complications	Symptomatic complications	Type of cement leakage
						16) Intravenous: 68,8% (11/16)
43. Reyad, RM et al.	Vertebroplasty * 63 patients * 92 vertebrae	Fluoroscopy * CT 1 week post-op	1,5 mL / vertebra	**Total: 82,5%** * Transient exacerbation of pain: 61,9% (39/63) * Local haematoma: 9,5% (6/63) * Cement leakage: 7,9% (5/63) * Adjacent level fracture: 3,2% (2/63)	**Total: 74,6%** * Exacerbation of pain: 61,9% *Local haematoma: 9,5% * Adjacent fracture: 3,2%	Paravertebral soft tissues: 20% (1/5) Intracanal: 20% (1/5) Intradiscal: 40% (2/5) Intravenous: 40% (2/5)
44. Saracen, A et al	Vertebroplasty * 90 patients * 137 vertebrae	Fluoroscopy * No post-op CT	2–3 mL / vertebra	**Total: 35,1%** * Cement leakage: 34% (47/137) * Pulmonary embolism: 1,1% (1/90)	**Total: 1,1%** * Monoparesis due to cement intracanal leakage, resolved with surgery: 1,1% (1/90)	Paravertebral soft tissue: 25,5% (12/47) Intracanal: 14,9% (7/47) Intradiscal: 19,2% (9/47) Intravenous: 40,4% (19/47)
45. Shi X et al.	Vertebroplasty * 309 patients	- * -	–	**Total: 31,2%** * Cement leakage: 31,2% (189/309)	**Total: 1%** * Neurological symptoms due to cement leakage: 1% (3/309)	Paravertebral soft tissue: 29,4% (74/252) Intracanal: 13,1% (33/252) Intradiscal: 22,6% (57/252) Intravenous: 34,9% (88/252) ** 252 types of leakage due to more than 1 type in the same vertebra*
46. Shimony, JS et al.	Vertebroplasty * 50 patients * 129 vertebrae	Fluoroscopy * -	–	**Total: 14%** * Increased or new sites of pain: 14% (7/50) ** Asymptomatic cement leakage was not considered as a complication*	**Total: 14%** * Increased or new sites of pain: 14% (7/50)	–
47. Tekin, SB et al.	Vertebroplasty & Kyphoplasty * 42 patients * 76 vertebrae	Fluoroscopy * No post-op CT	–	**Total: 7,1%** * Vacuum phenomena: 4,7% (2/42) * Cement leakage: 2,4% (1/42)	**0%**	–
48. Tian, Q et al.	Vertebroplasty * 39 patients * 51 vertebrae	Fluoroscopy * CT post-op	3,3 mL / vertebra (2–5 mL)	**Total: 38,5%** * Cement leakage: 38,5% (15/39)	**0%**	Paravertebral soft tissues: 60% (9/15) Intracanal: 6,7% (1/15) Intravenous: 20% (3/15) Puncture site: 13,3% (2/15)
49. Trumm, CG et al.	Vertebroplasty * 76 vertebrae	Fluoroscopy * CT post-op	3–3,5 mL / vertebra	**Total: 75%** * Cement leakage: 75% (57/76)	**0%**	Paravertebral soft tissue: (42,2% 27/64) Intracanal: 26,6% (17/64) Intradiscal: 31,2% (20/64)
50. Vrionis, FR et al	Kyphoplasty * 50 patients * 128 vertebrae	Fluoroscopy * No post-op CT	–	**Total: 2%** * Asystole: 2%	**Total: 2%** * Asystole: 2% *** successfully reversed, no cause found*	–
51. Wang, H et al.	Kyphoplasty * 36 patients * 36 vertebrae	- * No post-op CT	–	**Total: 11,1%** Cement leakage: 11,1% (4/36)	**0%**	Paravertebral soft tissue: 50% (2/4) Intradiscal: 25% (1/4) Intravenous: 25% (1/4)
52. Wang, L et al.	Vertebroplasty * 64 patients * 113 vertebrae	Fluoroscopy * CT post-op	4,84 mL / vertebra (1,2–9 mL)	**Total: 56,6%** * Cement leakage: 56,6% (64/113)	**0%**	Paravertebral soft tissue or intradiscal: 54,7% (35/64) Intracanal: 28,1% (18/64) Intravenous: 17,2% (11/64)
53. Wang, Y et al.	Kyphoplasty * 41 patients * 73 vertebrae	Fluoroscopy * CT post-op	3,6 mL / vertebra (1,5–6,5 mL)	**Total: 17,1%** * Cement leakage: 17,1% (7/41)	**0%**	–
54. Weill, A et al.	Vertebroplasty * 22 sessions	Fluoroscopy * CT post-op	–	**Total: 45,4%** * Cement leakage: 40,9% (9/	**Total: 9%** * Acute raquialgia because of	Paravertebral soft tissue: 11,1 (1/9)

(*continued on next page*)

Table 3 (*continued*)

Study	Procedure details	Imaging details	Cement details	Complications	Symptomatic complications	Type of cement leakage
	* 21 patients * 27 vertebrae			22) * Cement embolism to IVC: 4,5% (1/22)	cement leakage into canal: 9% (2/22) ** *need for surgical treatment in 1/2 cases*	Intracanal: 22,2 (2/9) Intradiscal: 11,1 (1/9) Intravenous: 55,6 (5/9)
55. Wu, W et al.	Kyphoplasty * 31 patients * 58 vertebrae	Fluoroscopy * CT post-op	2,6 mL / vertebra	**Total: 6,5%** * Cement leakage: 6,5% (2/31)	**0%**	Intradiscal: 50% (1/2) Puncture Site: 50% (1/2)
56. Wu, W et al.	Kyphoplasty * 59 patients * 59 vertebrae	Fluoroscopic * CT post-op	3,84 mL	**Total: 18,6%** * Cement leakage: 18,6% (11/59)	**0%**	Intracanal: 18,2% (2/11) Intradiscal: 36,4% (4/11) Intravenous: 18,2% (2/11) Puncture Site: 27,2% (3/11)
57. Yamada, K et al.	Vertebroplasty * 4 patients * 5 vertebrae	CT-guided * CT post-op	4 mL /vertebra (1,5–7,5 mL)	**Total: 50%** * Cement leakage: 50% (2/4)	**0%**	–
58. Yan, B C et al.	Vertebroplasty * 87 patients * 111 vertebrae	Fluoroscopy * CT post-op	5,87 mL/ vertebra	**Total: 114%** * Cement leakage: 98% (109/111) * Fever: 12,6% (11/87) * Pain: 3,4% (3/87)	**Total: 16%** * Fever: 12,6% (11/87) * Pain: 3,4% (3/87)	Paravertebral soft tissue: 49,1% (77/157) Intracanal: 20,4% (32/157) Intradiscal: 19,7% (31/157) Intravenous: 10,8% (17/157)
59. Zhang, H et al.	Kyphoplasty * 13 patients * 13 vertebrae	Fluoroscopy * No post-op CT	–	**Total: 15,4%** * Cement leakage: 15,4% (2/13)	**0%**	Intracanal: 33,3% (1/3) Intravenous: 66,7% (2/3)
60. Zhang, J et al.	Kyphoplasty * 28 patients * 37 vertebrae	Fluoroscopy * No post-op CT	3,70 mL / vertebra	**0%**	**0%**	–

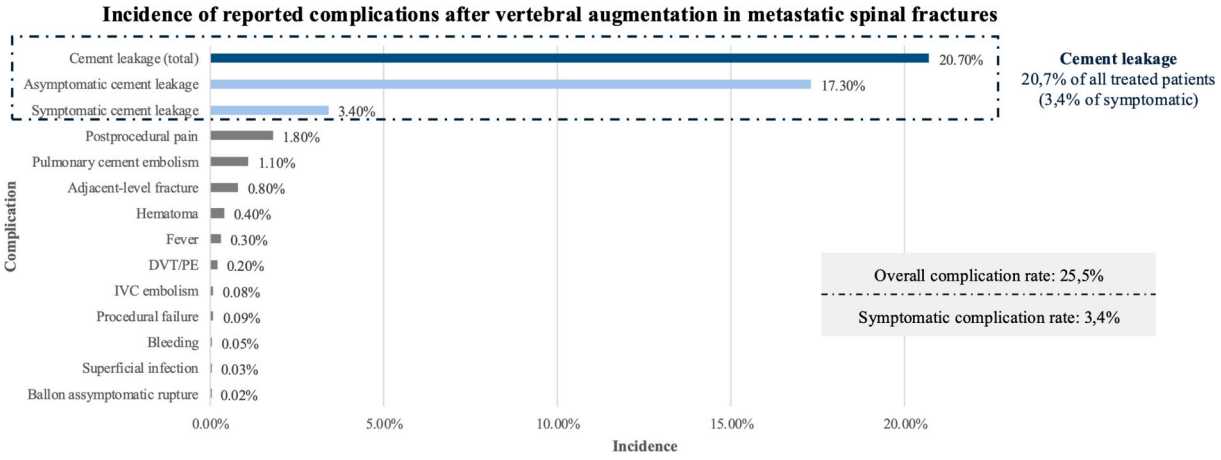


Fig. 2. Incidence of reported complications after verteble augmentation for metastatic spinal fractures. Values represent weighted inceidence(%) across the included studies, adjusted for study sample size. A patient may have experienced more than one complication.

significant leakage when reporting outcomes.

Experience with vertebral augmentation for osteoporotic and traumatic fractures may inform the interpretation of findings in metastatic disease. In osteoporotic fractures, systematic reviews report cement leakage rates of 18.4% to 54.7%, depending on the augmentation technique, with most events radiographic rather than clinically significant [91,92]. These rates are comparable to, or even higher than, the weighted leakage rate of 20.7% observed in our metastatic population. Importantly, cortical disruption has been identified as a major risk factor for leakage in osteoporotic fractures, together with intravertebral clefts, lower cement viscosity, and greater cement volume [92]. These factors may also be relevant in metastatic lesions, particularly when cortical destruction is present; however, the heterogeneity of the included studies precluded their assessment in our review. Evidence for traumatic fractures is more limited but shows a similar predominance of radiographic over symptomatic leakage: a systematic review identified cement leakage as the most reported complication, without associated clinical consequences [93], whereas a CT-based series detected leakage in 50% of cases, all asymptomatic [94]. Although these findings may suggest a comparable pattern of procedural safety, the risk and spectrum of complications cannot be fully extrapolated across indications, given their distinct biological settings. Metastatic disease may be associated with complications not described in the osteoporotic or traumatic literature, such as tumour extrusion and needle-tract tumour seeding, as described in this review.

4.2. Pulmonary cement embolism

Pulmonary cement embolism following venous leakage has been linked to treatment of multiple vertebral levels, involvement of the thoracic vertebrae, and use of larger cement volumes or lower cement viscosity [11,29,45]. Similar factors have been identified in a systematic review and meta-analysis of predominantly osteoporotic vertebral fractures, which reported an overall PCE incidence of 2.3% [95]. In our review, we identified an overall incidence of 1.08%. This overlap in reported risk factors suggests that cement embolisation may be substantially influenced by procedural factors rather than by the underlying vertebral pathology alone. Nevertheless, the influence of the underlying disease on the clinical significance of PCE remains unclear. Although some authors describe it as an incidental finding on CT scans, several symptomatic cases have been reported, not only in case reports but also in observational studies (accounting for 4% of all symptomatic complications). Therefore, although the true incidence of cement embolism after venous leakage remains unclear, it is important to be aware of its potential in symptomatic patients.

4.3. Influence of multilevel vertebral augmentation

Multilevel vertebral augmentation has been associated with greater complexity, leading to increased procedural difficulty and risks, including cement leakage and embolic complications [34]. Although treatment of more than 3 vertebrae in a single session occurred in 47,3% of studies, these cases represented a minority of the patients within each study, and there was no significant correlation with increased complications ($p = 0.45$). However, the absence of subgroup analysis in studies involving multilevel procedures affects the ability to draw definitive conclusions, and further research is necessary to better understand the safety of such procedures.

4.4. Influence of type of procedure

Some studies have suggested that the type of augmentation procedure may influence complication rates, with lower rates of cement leakage with kyphoplasty [11,46]. However, in this review, there was no statistically significant association between procedure type and complication occurrence ($p = 0.27$).

4.5. Influence of CT evaluation

Postoperative imaging protocols may also affect the reported rates of complications. In studies in which patients underwent intraoperative or postoperative CT evaluation, the rate of detected cement leakage was significantly higher, but not the rate of symptomatic complications. This reflects CT's higher sensitivity for detecting small cement leaks [16], but its clinical relevance is unknown. Further studies are needed to determine whether the increased sensitivity of CT justifies its cost and routine use, given that most detected cement leaks are asymptomatic.

4.6. Rare complications

The case reports provided valuable information about rare complications that are not commonly described in larger observational studies. Some of these complications, such as neurological compromise from cement leakage and symptomatic pulmonary cement embolisms, are described in the included observational studies, but extremely rare complications, including needle-tract seeding, tumour extrusion, and cement migration into cardiac structures, were reported almost exclusively in case reports. These findings emphasise that although most complications linked to vertebral augmentation procedures are rare, infrequent yet potentially serious events can still occur and should always be considered if such symptoms arise.

Beyond documented clinical complications, Mohme et al. reported a transient increase in circulating tumour cells after cement augmentation of vertebral metastases, raising the possibility of procedure-related mechanical dissemination of tumour cells. Although the clinical significance of this finding remains uncertain, it highlights a potential biological effect of vertebral augmentation that warrants further investigation [96].

DVT and non-cement pulmonary embolism were also reported as rare postoperative events. However, their direct relationship with vertebral augmentation remains uncertain, as patients with metastatic disease have multiple concurrent risk factors for venous thromboembolism, including malignancy and reduced mobility. This further supports the need for a clear definition of postoperative complications, as these events may be related to the underlying disease or the peri-procedural setting rather than to cement injection itself.

4.7. Strengths and limitations

This review also has some limitations. First, the heterogeneity of study designs and reporting methods limited comparison between studies and the ability to draw firm conclusions about complication rates and risk factors. In addition, many of the included studies were case series or retrospective observational studies, which may be affected by selection bias and incomplete reporting of complications. Furthermore, the definition and reporting of complications varied widely across studies, particularly for asymptomatic cement leakage, which was not always considered a complication. Another factor that was not consistently reported is follow-up duration; when available, it varied considerably. This variability may influence the detection of complications, particularly events such as adjacent-level fractures that can occur only months after the procedure. Furthermore, the radiological pattern of metastatic bone involvement (lytic, sclerotic, or mixed), patient-selection criteria, and prior oncological treatments were inconsistently reported, preventing assessment of their potential influence on complication rates. Thus, despite the large number of available studies, the overall quality of evidence remains limited, highlighting the need for higher-quality prospective studies rather than simply a greater volume of literature. Additionally, studies combining vertebral augmentation with other spinal interventions, such as surgical fixation or tumour ablation, were excluded to minimise confounding, as reported complications could not be reliably attributed to vertebral augmentation alone.

Despite these limitations, this review also has several strengths. First, it includes many patients and treated vertebral levels across the analysed studies, with more than 3000 patients and over 5700 treated vertebrae, providing a broad overview of complications associated with vertebral augmentation procedures in metastatic spinal fractures. In addition, by including different study designs, it was possible to identify both common and rare complications. To our knowledge, few previous reviews have attempted to systematically describe the full spectrum of complications, including uncommon but clinically relevant events such as needle tract tumour seeding, tumour extrusion, and intracardiac cement embolism. This broader perspective may help provide a more complete understanding of the safety profile of vertebral augmentation in oncologic patients.

5. Conclusion

The available evidence suggests that vertebral augmentation is a safe treatment option for painful metastatic vertebral fractures, with a low rate of symptomatic complications (< 4%). However, important uncertainties remain, particularly the role of multilevel procedures, the clinical significance of cement leakage, and the relevance of CT evaluation. The apparent safety profile of vertebral augmentation depends strongly on how complications are defined and detected. A clearer understanding of procedural safety is essential to support informed clinical decision-making, optimise procedural planning, and appropriately balance potential benefits and risks in this vulnerable patient population.

CRediT authorship contribution statement

Maria Clara Correia: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Ana Margarida Vieira:** Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Filipe Duarte:** Writing – review & editing, Validation, Supervision, Resources, Formal analysis, Data curation, Conceptualization. **André Rodrigues Pinho:** Writing – review & editing, Project administration, Methodology, Investigation, Conceptualization. **Francisco Serdoura:** Supervision. **Maria João Leite:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Investigation, Formal analysis, Conceptualization.

Ethics approval

Not applicable. This study is a systematic review of previously published data.

Consent to participate.

Not applicable.

Consent for publication.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

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